Mehak Khan

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Technical Skills

Analog & Mixed-Signal: Cadence Virtuoso, Xschem, ngspice, SKY130 PDK, SPICE, Verilog-A, g_m/I_D analysis
PCB & Embedded: KiCad, Altium, STM32, ESP32-S3, I2C/SPI/PWM, Oscilloscope, Logic & Spectrum Analyzers
Digital & Programming: SystemVerilog, AMD Vivado/Vitis, RISC-V, Python, C++, PyTorch, snnTorch

Experience

Electrical & Instrumentation Commissioning Intern — Jan 2026 – Apr 2026
Technip Energies France — *Ras Laffan Industrial City, Qatar*

- Validated grid–generation changeover and restoration logic across a multi-source **GTG/STG/EDG + utility-grid** network supporting a major LNG facility.
- Performed secondary injection and functional testing on **132kV/66kV GIS protection relays**, including 50/51, 87, and 74TC protection functions.
- Characterized industrial **online double-conversion UPS systems**, analyzing SCR rectifiers, IGBT inverter stages, DC-link behavior, LC filtering, harmonics, ripple, and EMI.
- Verified **IEC 61850 and Modbus TCP/IP** communication across MV/HV switchgear, RTUs, and control systems while supporting validation of **200+ LSS/PMS I/O signals**.
- Audited GPS/IRIG-B time synchronization and grounding schemes supporting high-resolution fault analysis and coordinated protection workflows.

Software Developer Intern — Sep 2024 – Dec 2024
Toyota Canada Inc. — *Scarborough, ON*

- Built Python/AWS automation and analytics systems that reduced infrastructure update time by **60%**, automated **70% of manual IT patching**, and supported $100K+ in monthly financial reporting.

Backend Developer Intern — Jan 2024 – Apr 2024
Landscape Direct Corps. (Startup) — *Carleton, ON*

- Built AWS/Ruby on Rails backend systems, reducing database query latency by **35%** and automating workflows that saved **15+ hours/week**.

Projects

PDK-Agnostic Subthreshold Voltage Reference | *SKY130, Xschem, ngspice, Verilog-A*

- Designed a **MOSFET-only subthreshold voltage reference** targeting temperature-stable operation from approximately $-40°$C to 120°C without relying on process-specific bipolar devices.
- Developed and validated a **Verilog-A behavioral model** against transistor-level simulation and characterized device biasing using **g_m/I_D analysis** for Tiny Tapeout integration.

Neuromorphic Stress-Responsive Wearable | *PyTorch, snnTorch, SKY130, ngspice*

- Developed a **4-bit quantized spiking neural network** for ECG/EDA-derived physiological stress classification, achieving **82.3% balanced accuracy** and **82.0% stress recall** in Phase-1 development testing.
- Mapped trained network weights to **measured SKY130 NMOS conductances and gate biases** and demonstrated transistor-level two-cell analog current summation in ngspice.

Watolink – Hardware Lead | *ESP32-S3, KiCad/Altium, Mixed-Signal PCB*

- Led hardware architecture and schematic/PCB design for a wearable embedded system integrating **ESP32-S3, ADS1115 ADC, force sensors, and multi-channel PWM actuation**.
- Designed the mixed-signal power-delivery architecture and evaluated component/manufacturability tradeoffs, including a **BQ24040 power-management redesign**, grounding strategy, and analog/digital partitioning.

Pipelined FPGA Matrix-Vector Accelerator | *SystemVerilog, Vivado, Kria KV260*

- Designed a parameterized MVM accelerator with SRAM-backed storage, parallel output lanes, per-lane accumulation, and **FSM-controlled address sequencing**.
- Implemented a **6-cycle pipelined 8-element dot-product engine** with eight parallel multipliers and a balanced $4 \to 2 \to 1$ reduction tree; recorded a **2.621 ns post-implementation critical datapath**.

Education

University of Waterloo — Waterloo, ON
BASc in Electrical Engineering (Honours), President's Scholarship of Distinction — *Sep 2023 – Apr 2028*